Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]

[20EE31]

THIRD SEMESTER B.Tech EXAMINATION, MARCH - 2022

ELECTRICAL CIRCUIT ANALYSIS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1. **a.** Obtain the equivalent resistance R_{ab} for the circuit in Fig. 1. and use it to find current i. **07**

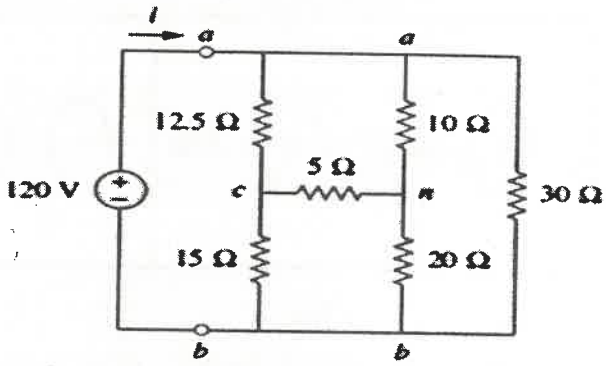

Fig. 1.

b. A loudspeaker is connected to an amplifier as shown in Fig. 2. If a 10 Ω loudspeaker draws the maximum power of 12 W from the amplifier, determine the maximum power a 4 Ω loudspeaker will draw. **04**

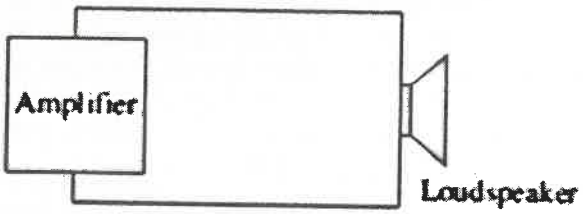

Fig. 2.

c. For the circuit in Fig. 3, find the branch currents I_1, I_2 and I_3 using mesh analysis. **03**

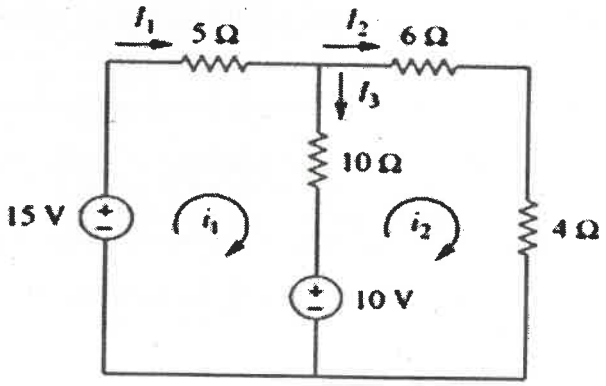

Fig. 3.

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2. a. Obtain the dual of the circuit in Fig. 4 and Fig. 5.

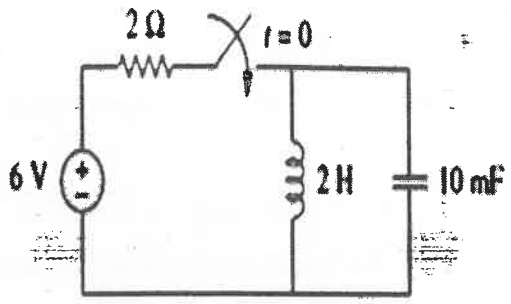

Fig. 4.

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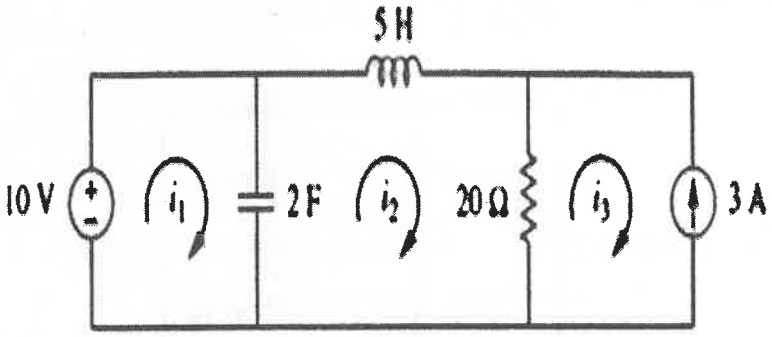

Fig. 5.

b. Define the time constants of RL and RC Circuits. What is its significance. 04

3. a. An electrical load operates at 240V rms. The load absorbs an average power of 8 kW at a lagging power factor of 0.8. Calculate, the impedance and complex power of the load. 06

b. Two identical coils when connected in series across a source, the total inductance is found to be 3.0 H. When one of the coils is reversed the total inductance become 1.4 H. Find the mutual inductance between them and the self-inductance of each coil, Also find the coupling coefficient. 08

OR

4. a. A star connected impedance draws a power of 5 kW and a leading current of 100 A from a three phase 415 V, 50 Hz, R,Y,B phase sequence balanced supply. Find the impedance and the circuit elements. Also find the reactive power and power factor. 06

b. An unbalanced delta connected load of $Z_1 = (10 - j10)\,\Omega$, $Z_2 = (20 - j20)\,\Omega$ and $Z_3 = (25 - j25)\,\Omega$ are connected across a balanced three phase supply of 415 V, 50 Hz. Find the phase and line currents and the total active power drawn by this load. 05

c. Determine the rms value of the current waveform in Fig. 7. If the current is passed through a resistor, find the average power absorbed by the 2 Ω resistor. 03

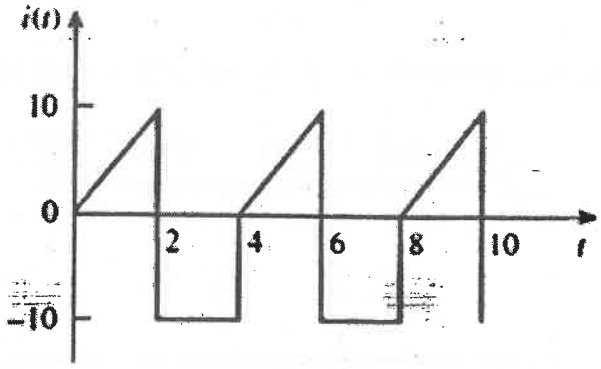

Fig. 7.

5. **a.** An RL series circuit is energized by a supply of 10 V DC by closing a switch in series with it. $R = 10\,\Omega$ and $L = 5\,mH$. The circuit was under steady state before closing the switch. Find the current for time 't' greater than zero. What is the current after 5 milli seconds? Use Laplace transformations. **06**

b. Calculate the Laplace transform of the periodic function in Fig. 8. **04**

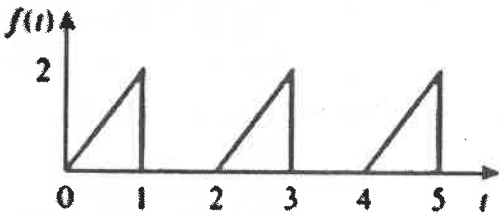

Fig. 8.

c. What is an impulse function? Show that the Laplace transformation of a unit impulse function is unity? **04**

OR

6. **a.** An RLC series circuit has a capacitance of 320 pico Farad connected across a source of 0.85 V has Q of 50 and resonant frequency of 175 kHz. Find the inductance, resistance, the current and the bandwidth of the circuit. **08**

b. Calculate $v(t)$ given that $V(s) = \frac{10\,s^2+4}{s(s+1)(s+2)^2}$ **06**

7. **a.** Define z and y parameters and obtain the relationship between them. Draw the equivalent circuit of Z – parameters. **06**

b. Determine the z parameters for the circuit in Fig. 9. **04**

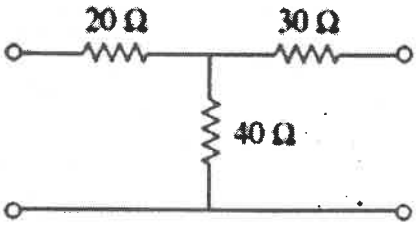

Fig. 9.

c. Find $[z]$ and $[g]$ of a two-port network if **04**

$$[T] = \begin{bmatrix} 10 & 1.5\,\Omega \\ 2\,S & 4 \end{bmatrix}$$

OR

8. **a.** Determine the hybrid parameters for the network in Fig. 10. **06**

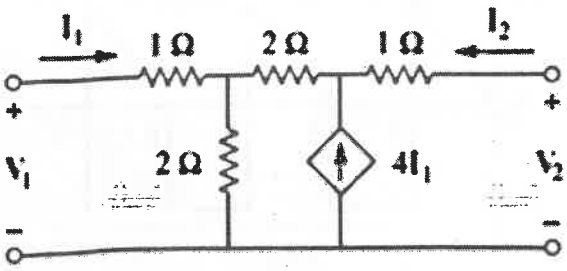

Fig. 10.

b. Determine the *z* and *y* parameters for the circuit in Fig. 11. **04**

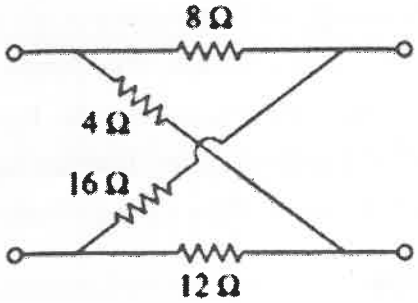

Fig. 11.

c. Find the **ABCD** parameters for the circuit in Fig. 12. **04**

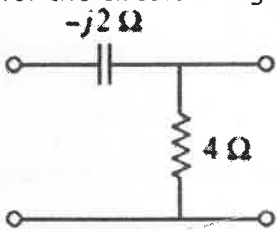

Fig. 12.